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Subject = oop

Lab task = polymorphism

<u>Question NO = 01: simple polymorphism code:</u>

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Exe.java

```java
class Shape     // Babar Zareen Sap ID 74760
{
    void draw(){
        System.out.println("Shape........");
    }
}
class Ractangle extends Shape
{
    void draw(){
        System.out.println("Ractangle.......");
    }
}
class Circle extends Shape  // Babar Zareen Sap ID 74760
{
    void draw(){
        System.out.println("Circle........");
    }
}               // Babar Zareen Sap ID 74760

public class Exe{
    public static void main(String[] args) {
        Ractangle r = new Ractangle();
        r.draw();
    }
}
```

```
Ractangle.......


...Program finished with exit code 0
Press ENTER to exit console.
```

Question NO = 02: Method Overriding (Runtime Polymorphism) code:

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Overriding.java

```java
class Animal //Babar Zareen Sap ID 74760
{
    void sound() {
        System.out.println("Animal makes sound");
    }
}       //Babar Zareen Sap ID 74760

class Dog extends Animal
{
    void sound()
    {
        System.out.println("Dog barks");
    }
}       //Babar Zareen Sap ID 74760

public class Overriding
{
    public static void main(String[] args)
    {
        Dog d = new Dog();
        d.sound();
    }
}
```

```
Dog barks


...Program finished with exit code 0
Press ENTER to exit console.
```

Question NO = 03: Method Overloading (Compile Time Polymorphism)

code:

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Overloading.java

```java
class Calculator {  //Babar Zarren sap Id 74760
    int add(int a, int b)
    {
        return a + b;
    }

    int add(int a, int b, int c)
    {
        return a + b + c;
    }       //Babar Zarren sap Id 74760

    double add(double a, double b)
    {
        return a + b;
    }    //Babar Zarren sap Id 74760
}
public class Overloading {
    public static void main(String[] args) {
        Calculator c = new Calculator();

        System.out.println(c.add(5, 3));
        System.out.println(c.add(5, 3, 2));
        System.out.println(c.add(5.5, 3.5));
    }
}
```

```
8
10
9.0


...Program finished with exit code 0
```